

Qualification Pack



Research Associate- Pharma and Biological Products

API Synthesis & Medicinal Chemistry/ Formulation Development/ Biological Product Development / Technology translation and Process Development/ Technology transfer/ Analytical - Mass Spectrometry/ Analytical - NMR

Innovation driven Entrepreneurship

QP Code: LFS/Q0514

Version: 1.0

NSQF Level: 5.5



Qualification Pack

Life Sciences Sector Skill Development Council || # 14, Rear 2nd Floor, Palam Marg, Vasant Vihar
New Delhi-110057 || email:SHIVI.CHAUDHARY@LSSSDC.IN



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LFS/Q0514: Research Associate- Pharma and Biological Products

Brief Job Description

Research Associate- Pharma and Biological Products is responsible for assisting in biological product development, API synthesis, medicinal chemistry-based research, carrying out research related activities along with also assists in technology transfer and process development activities to large scale manufacturing. The jobholder is also responsible for reporting and documentation, problem solving and decision-making, Co-ordinate with manager and team members and maintaining healthy and safe working environment.

Personal Attributes

The individual should have good communication and interpersonal skills. The person should possess investigational abilities, analytical and reasoning skills. The role holder should have critical thinking skills along with excellent organizational skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [LFS/N0111: Ensure adherence to Environment, health and safety guidelines in production facility and GMP controlled areas by self and subordinates](#)
2. [LFS/N0129: Coordinate and communicate with supervisor/production chemist, research teams and auditors](#)
3. [LFS/N0572: Maintain experiment records ensuring data integrity and intellectual property rights](#)
4. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: API Synthesis & Medicinal Chemistry

1. [LFS/N0516: Assist in development of new Active Pharmaceutical Ingredient \(API\) / Synthetic Product Development](#)
2. [LFS/N0573: Assist in Medicinal Chemistry research](#)

Elective 2: Formulation Development

1. [LFS/N0574: Assist in new Non Sterile formulation product development](#)



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2. [LFS/N0575: Assist in aseptic product development](#)

Elective 3: Biological Product Development

1. [LFS/N0576: Assist in new biological product development](#)

Elective 4: Technology translation and Process Development

1. [LFS/N0517: Assist in technology translation and process development](#)

Elective 5: Technology transfer

1. [LFS/N0577: Assist in technology transfer](#)

Elective 6: Analytical - Mass Spectrometry

1. [LFS/N0578: Conducts research and analysis using mass spectrometry techniques to solve complex problems, develop new products, or improve existing ones.](#)

Elective 7: Analytical - NMR

1. [LFS/N0579: Conducts research and analysis using Nuclear Magnetic Resonance techniques to solve complex problems, develop new products, or improve existing ones.](#)

Options(Not mandatory):

Option : Innovation driven Entrepreneurship

1. [LFS/N0126: Establish innovation led enterprise and perform various entrepreneurial activities to run the regulated business operations](#)
2. [LFS/N0127: Maintain the Critical Business documents as Innovation led entrepreneur](#)
3. [LFS/N0128: Product commercialization and Intellectual Property Rights Management](#)

Qualification Pack (QP) Parameters

Sector	Life Sciences
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Sub-Sector	Pharmaceutical, Bio Pharmaceutical
Occupation	Research and Development
Country	India
NSQF Level	5.5
Credits	57
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2131.1300
Minimum Educational Qualification & Experience	Pursuing 2nd year of PG (after 3 year UG Degree) (M.Sc in relevant filed) OR Pursuing 2nd year of PG (after 3 year UG Degree) (M.Pharma) OR Pursuing 2nd year of PG (after 3 year UG Degree) (M.Tech in relevant Engineering domain)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	NA
Next Review Date	29/09/2026
NSQC Approval Date	29/09/2023
Version	1.0
Reference code on NQR	QG-5.5-LS-00997-2023-V1-LSSSDC
NQR Version	1.0

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LFS/N0111: Ensure adherence to Environment, health and safety guidelines in production facility and GMP controlled areas by self and subordinates

Description

This job function is about the job role holder ensuring adherence to the health, hygiene, safety and environment guidelines while working in the production facility and GMP controlled areas by self and subordinates

Scope

The scope covers the following :

- Follow health and hygiene protocols
- Adherence to safety and security procedures
- Adherence to emergency procedures

Elements and Performance Criteria

Follow health and hygiene protocols

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and personal hygiene-related protocols as per WHO standards and ICH GMP guidelines
- PC2.** wash hands before entering in the production area with soap/alcohol based sanitisers
- PC3.** report any allergy, sickness or any other environment-related breach before or after entering the work premises to the designated person
- PC4.** take preventive actions on the report of any allergy, sickness or any other environment-related breach reported by subordinates
- PC5.** follow gowning procedures while entering an environment controlled work area and ensure adherence to the same by others

Adherence to safety and security procedures

To be competent, the user/individual on the job must be able to:

- PC6.** comply with safety and security policies and procedures
- PC7.** ensure the use of appropriate safety gears like headgear, masks, gloves and other accessories as mentioned in the guidelines, by self and subordinates while carrying out work
- PC8.** take preventive and corrective actions based on the report of any identified breaches in safety and security policies and procedures by subordinates
- PC9.** ensure that discipline for material segregation and 5S system is followed at the storage area
- PC10.** comply with material handling, segregation, and storage guidelines for hazardous material
- PC11.** take corrective actions for reported hazards in consultation with EHS personnel
- PC12.** complete the records of safety drills and trainings undertaken by self and subordinates

Adherence to emergency procedures

To be competent, the user/individual on the job must be able to:

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- PC13.** report any hazards that he/she is not competent to deal with the relevant EHS personnel and warn other people who may be affected
- PC14.** raise the alarm and inform the concerned person immediately for action in the cases of spill, fall, injury, toxic inhale, fire or explosion
- PC15.** follow emergency protocols for any alarms and ensure the safety of subordinates in the area under supervision
- PC16.** follow emergency procedures efficiently
- PC17.** ensure injured employees are provided appropriate first aid and medical aid

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's procedures for the environment, health, and safety
- KU2.** implications that any non-compliance with health, safety and security may have on individuals and the organization
- KU3.** workplace hazards in the manufacturing facility in the life sciences sector, how and when to report hazards
- KU4.** limits of individual responsibility for dealing with hazards
- KU5.** chemical substances, their characteristics, and required precaution and safety measures
- KU6.** gowning procedure
- KU7.** the organization's emergency procedures for different emergencies and the importance of following these
- KU8.** evacuation procedures for employees, contract staff and visitors
- KU9.** how to summon medical assistance and the emergency services, where necessary
- KU10.** health, safety and accident reporting procedures and the importance of reporting in GMP
- KU11.** different types of breaches in the environment, health, safety and security and how and when to report these
- KU12.** WHO guidelines for personal hygiene
- KU13.** type of safety gears and procedure to use them
- KU14.** the importance of material segregation and 5S system
- KU15.** WHO guidelines for handling and storing hazardous material

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language



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- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS6.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS8.** use verbal communication skills to communicate with supervisor/ manager/ EHS Incharge or any other concerned authority clearly for escalating any emergency or hazard

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and hygiene protocols</i>	10	15	-	5
PC1. comply with health and personal hygiene-related protocols as per WHO standards and ICH GMP guidelines	-	-	-	-
PC2. wash hands before entering in the production area with soap/alcohol based sanitisers	-	-	-	-
PC3. report any allergy, sickness or any other environment-related breach before or after entering the work premises to the designated person	-	-	-	-
PC4. take preventive actions on the report of any allergy, sickness or any other environment-related breach reported by subordinates	-	-	-	-
PC5. follow gowning procedures while entering an environment controlled work area and ensure adherence to the same by others	-	-	-	-
<i>Adherence to safety and security procedures</i>	10	25	-	5
PC6. comply with safety and security policies and procedures	-	-	-	-
PC7. ensure the use of appropriate safety gears like headgear, masks, gloves and other accessories as mentioned in the guidelines, by self and subordinates while carrying out work	-	-	-	-
PC8. take preventive and corrective actions based on the report of any identified breaches in safety and security policies and procedures by subordinates	-	-	-	-
PC9. ensure that discipline for material segregation and 5S system is followed at the storage area	-	-	-	-
PC10. comply with material handling, segregation, and storage guidelines for hazardous material	-	-	-	-
PC11. take corrective actions for reported hazards in consultation with EHS personnel	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. complete the records of safety drills and trainings undertaken by self and subordinates	-	-	-	-
<i>Adherence to emergency procedures</i>	10	15	-	5
PC13. report any hazards that he/she is not competent to deal with the relevant EHS personnel and warn other people who may be affected	-	-	-	-
PC14. raise the alarm and inform the concerned person immediately for action in the cases of spill, fall, injury, toxic inhale, fire or explosion	-	-	-	-
PC15. follow emergency protocols for any alarms and ensure the safety of subordinates in the area under supervision	-	-	-	-
PC16. follow emergency procedures efficiently	-	-	-	-
PC17. ensure injured employees are provided appropriate first aid and medical aid	-	-	-	-
NOS Total	30	55	-	15



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0111
NOS Name	Ensure adherence to Environment, health and safety guidelines in production facility and GMP controlled areas by self and subordinates
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Contract Research
Occupation	Generic
NSQF Level	5
Credits	2.00
Version	2.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023



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LFS/N0129: Coordinate and communicate with supervisor/production chemist, research teams and auditors

Description

This NOS unit is about coordinating with supervisor/ production chemist, research teams, and auditors.

Scope

The scope covers the following :

- Coordination with supervisor/ production chemist
- Coordination with cross-functional teams
- Coordination with auditors
- Sensitivity towards all genders and people with disability

Elements and Performance Criteria

Coordination with Supervisor / production chemist

To be competent, the user/individual on the job must be able to:

- PC1.** work as per instructions given by reporting supervisor
- PC2.** seek guidance/advice from supervisor on production plan for meeting the timelines
- PC3.** communicate process-flow improvements, production defects received from the previous process, repairs and maintenance of equipment to reporting supervisor/ production chemist
- PC4.** ensure timely intimation to supervisor/ production chemist about planned absence/ illness/ dizziness during work/ critical issues requiring his/her intervention
- PC5.** coordinate with supervisor on work-related and behavioral feedback

Coordination with cross-functional teams

To be competent, the user/individual on the job must be able to:

- PC6.** support team members and colleagues of other departments in work
- PC7.** take handover from previous shift operator and give handover to next shift operator as per SOP
- PC8.** guide manufacturing and packaging assistants during production process
- PC9.** coordinate with warehouse team for material dispensing and issuance
- PC10.** coordinate with maintenance team for preventive and corrective maintenance, break down and calibration errors
- PC11.** coordinate with quality control team for sample collection and batch release
- PC12.** coordinate with QA for machine/ equipment validation at a routine interval as per SOP
- PC13.** provide inputs to the concerned stakeholders in periodic fence line review to detect non-compliance
- PC14.** coordinate with EHS team for any safety incident, accident and emergency

Coordination with auditors

To be competent, the user/individual on the job must be able to:

- PC15.** provide clear answers to the auditor's queries

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PC16. provide the required documents of performed activities and operations to auditors on time

PC17. maintain data integrity while responding to auditors and regulatory inspectors

Sensitivity towards all genders and people with disability

To be competent, the user/individual on the job must be able to:

PC18. respect all genders, religions, and caste

PC19. empathize with people with disability

PC20. offer support or help to a person with disability only when asked

PC21. ensure to adhere with the guidelines laid in Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act

PC22. report any violation of prevention of sexual harassment (POSH) rules immediately to the POSH committee

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the company's policies on the preferred language of communication, reporting and escalation policy

KU2. the reporting structure of the organization

KU3. types of audits in the life sciences sector for the manufacturing operations

KU4. the required regulatory and statutory compliance-related documentation

KU5. the guidelines for data integrity, ethics, and compliance in the life sciences industry

KU6. the guidelines laid on Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act

KU7. the methods of workplace communication

KU8. importance of team coordination

KU9. the types of possible disabilities among people with disability (PwD)

KU10. the challenges faced by PwD

KU11. the importance of displaying empathy towards PwD

KU12. the right way to use the laws, acts, and provisions defined for PwD by the statutory bodies

KU13. the importance of awareness for gender sensitization and prevention of sexual harassment (POSH) act

KU14. importance of respecting all gender identities, religion, caste, and culture

KU15. method to receive the performance feedback

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. use reading and comprehension skills to gauge the relevant information manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/ comments



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- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written or computer-based record/ electronic mail in a given format and compliant with ALCOA principle
- GS3.** use verbal communication skills to communicate confidential and sensitive information discretely to the authorized person while interacting with teammates
- GS4.** use team-building skills during the interaction with teammates while managing the difficult/stressful or emotional situations at work
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical thinking skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** apply customer-centricity skills while responding to auditors and QA personnel

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Coordination with Supervisor / production chemist</i>	10	10	5	5
PC1. work as per instructions given by reporting supervisor	-	-	-	-
PC2. seek guidance/advice from supervisor on production plan for meeting the timelines	-	-	-	-
PC3. communicate process-flow improvements, production defects received from the previous process, repairs and maintenance of equipment to reporting supervisor/ production chemist	-	-	-	-
PC4. ensure timely intimation to supervisor/ production chemist about planned absence/ illness/ dizziness during work/ critical issues requiring his/her intervention	-	-	-	-
PC5. coordinate with supervisor on work-related and behavioral feedback	-	-	-	-
<i>Coordination with cross-functional teams</i>	10	10	5	5
PC6. support team members and colleagues of other departments in work	-	-	-	-
PC7. take handover from previous shift operator and give handover to next shift operator as per SOP	-	-	-	-
PC8. guide manufacturing and packaging assistants during production process	-	-	-	-
PC9. coordinate with warehouse team for material dispensing and issuance	-	-	-	-
PC10. coordinate with maintenance team for preventive and corrective maintenance, break down and calibration errors	-	-	-	-
PC11. coordinate with quality control team for sample collection and batch release	-	-	-	-
PC12. coordinate with QA for machine/ equipment validation at a routine interval as per SOP	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. provide inputs to the concerned stakeholders in periodic fence line review to detect non-compliance	-	-	-	-
PC14. coordinate with EHS team for any safety incident, accident and emergency	-	-	-	-
<i>Coordination with auditors</i>	10	10	5	5
PC15. provide clear answers to the auditor's queries	-	-	-	-
PC16. provide the required documents of performed activities and operations to auditors on time	-	-	-	-
PC17. maintain data integrity while responding to auditors and regulatory inspectors	-	-	-	-
<i>Sensitivity towards all genders and people with disability</i>	2	3	2	3
PC18. respect all genders, religions, and caste	-	-	-	-
PC19. empathize with people with disability	-	-	-	-
PC20. offer support or help to a person with disability only when asked	-	-	-	-
PC21. ensure to adhere with the guidelines laid in Sexual Harassment of Women at Workplace(Prevention, Prohibition and Redressal) Act	-	-	-	-
PC22. report any violation of prevention of sexual harassment (POSH) rules immediately to the POSH committee	-	-	-	-
NOS Total	32	33	17	18



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0129
NOS Name	Coordinate and communicate with supervisor/production chemist, research teams and auditors
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Contract Research
Occupation	Generic
NSQF Level	4
Credits	3.00
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

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LFS/N0572: Maintain experiment records ensuring data integrity and intellectual property rights

Description

This Job function is about the job holder maintaining the experiment records ensuring data integrity and intellectual property rights

Scope

The scope covers the following :

- The Scope covers the following :
- Reporting and recording
- Data Integrity
- Intellectual Property

Elements and Performance Criteria

Reporting and recording

To be competent, the user/individual on the job must be able to:

- PC1.** review and update the test methods and procedures as per the schedule or when a regulatory requirement arises according to the written procedures
- PC2.** fill logbooks, worksheet (an analytical record), reference standard entries, calibration records, parameters of column, reagent, volumetric solution and working standards
- PC3.** report defects/problem/incidents/quality issues/test results as applicable in a timely manner to the appropriate authority as laid down by the company
- PC4.** prepare analytical reports for detailed findings and recommendations as per SOPs
- PC5.** maintain lab records in LNB and eLNB

Data Integrity

To be competent, the user/individual on the job must be able to:

- PC6.** document the results of the testing and analysis accurately
- PC7.** maintain all original and controlled document files and quality records in a timely and accurate manner following ALCOA PLUS principles
- PC8.** respond to requests for information in an appropriate manner whilst following organizational procedures
- PC9.** make sure documents are available to all appropriate authorities to inspect/ audit
- PC10.** maintain the confidentiality of the research projects and processes

Intellectual Property

To be competent, the user/individual on the job must be able to:

- PC11.** create a comprehensive report at asset level, detailing every item of content used within a product and the rights associated with it
- PC12.** daily monitoring, logging, prioritizing assessment requests entering them onto appropriate regional tracker.

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PC13. ensure that all titles and asset information is reviewed thoroughly and in-depth

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of relevant regulatory requirements and quality standards related to test methods and procedures.
- KU2.** how to review and update test methods and procedures according to established schedules and regulatory changes.
- KU3.** Proficiency in filling logbooks, worksheets, reference standard entries, calibration records, and maintaining records of parameters
- KU4.** Ability to prepare analytical reports that detail findings and recommendations in accordance with Standard Operating Procedures (SOPs).
- KU5.** how to accurately document the results of testing and analysis, following data integrity principles.
- KU6.** Understanding the significance of preserving the confidentiality of research projects and processes.
- KU7.** Knowledge of the daily monitoring, logging, and prioritization of assessment requests and their entry into the appropriate tracking system
- KU8.** Understanding the importance of a thorough and in-depth review of all titles and asset information.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS6.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS7.** apply problem-solving skills to find solutions for workflow-related difficulties

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Reporting and recording</i>	10	10	10	10
PC1. review and update the test methods and procedures as per the schedule or when a regulatory requirement arises according to the written procedures	-	-	-	-
PC2. fill logbooks, worksheet (an analytical record), reference standard entries, calibration records, parameters of column, reagent, volumetric solution and working standards	-	-	-	-
PC3. report defects/problem/incidents/quality issues/test results as applicable in a timely manner to the appropriate authority as laid down by the company	-	-	-	-
PC4. prepare analytical reports for detailed findings and recommendations as per SOPs	-	-	-	-
PC5. maintain lab records in LNB and eLNB	-	-	-	-
<i>Data Integrity</i>	10	10	10	10
PC6. document the results of the testing and analysis accurately	-	-	-	-
PC7. maintain all original and controlled document files and quality records in a timely and accurate manner following ALCOA PLUS principles	-	-	-	-
PC8. respond to requests for information in an appropriate manner whilst following organizational procedures	-	-	-	-
PC9. make sure documents are available to all appropriate authorities to inspect/ audit	-	-	-	-
PC10. maintain the confidentiality of the research projects and processes	-	-	-	-
<i>Intellectual Property</i>	5	5	5	5



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. create a comprehensive report at asset level, detailing every item of content used within a product and the rights associated with it	-	-	-	-
PC12. daily monitoring, logging, prioritizing assessment requests entering them onto appropriate regional tracker.	-	-	-	-
PC13. ensure that all titles and asset information is reviewed thoroughly and in-depth	-	-	-	-
NOS Total	25	25	25	25



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0572
NOS Name	Maintain experiment records ensuring data integrity and intellectual property rights
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical
Occupation	Research and Development
NSQF Level	5.5
Credits	3.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023



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DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

Qualification Pack

- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



Qualification Pack

- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

LFS/N0516: Assist in development of new Active Pharmaceutical Ingredient (API) / Synthetic Product Development

Description

This NOS unit is about the the job role holder assisting in development of new Active Pharmaceutical Ingredient (API) / Synthetic Product Development

Scope

The scope covers the following :

- The NOS Covers the following:
- Pre product development activities
- API/synthetic product development activities

Elements and Performance Criteria

Pre product development activities

To be competent, the user/individual on the job must be able to:

- PC1.** support in developing exploratory development proposals for new innovation products
- PC2.** perform a literature search for the new product development
- PC3.** support in carrying out technical feasibility analysis on the NPD ideas
- PC4.** collect and analyze current development plans, new scientific developments and competitive activities
- PC5.** carryout research on assigned research problems contributing to the attainment of a defined research objective

API/synthetic product development activities

To be competent, the user/individual on the job must be able to:

- PC6.** supporting in execution of product (innovation) development related activities
- PC7.** conduct stability studies, generate stability data & prepare stability reports for innovation products
- PC8.** support execution of allocated product development responsibilities
- PC9.** perform standardization, optimization and scale-up of API synthesis reaction
- PC10.** monitor reaction work-up and extraction process for API synthesis
- PC11.** monitor and conduct regular checks on equipment and instrument conditions, document calibrations
- PC12.** use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required
- PC13.** perform all the experimentation in adherence to safety, GRP and GLP guidelines
- PC14.** identify issues on projects and report to higher authorities
- PC15.** assist in resolution of issues supported by good data analysis

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Understanding of the organization's strategic goals and market trends.
- KU2.** Familiarity with relevant scientific journals and publications.
- KU3.** Knowledge of technical feasibility assessment methods.
- KU4.** Understanding of intellectual property and competitive landscape
- KU5.** Knowledge of product development processes.
- KU6.** Knowledge of stability testing protocols and guidelines
- KU7.** Understanding of chemical processes and scale-up factors
- KU8.** Knowledge of chemical separation techniques.
- KU9.** Understanding of instrument operation and calibration standards.
- KU10.** Knowledge of Good Laboratory Practices (GLP) and Good Research Practices (GRP).
- KU11.** Understanding of escalation procedures and communication channels.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre product development activities</i>	20	10	10	10
PC1. support in developing exploratory development proposals for new innovation products	-	-	-	-
PC2. perform a literature search for the new product development	-	-	-	-
PC3. support in carrying out technical feasibility analysis on the NPD ideas	-	-	-	-
PC4. collect and analyze current development plans, new scientific developments and competitive activities	-	-	-	-
PC5. carryout research on assigned research problems contributing to the attainment of a defined research objective	-	-	-	-
<i>API/synthetic product development activities</i>	20	6	10	10
PC6. supporting in execution of product (innovation) development related activities	-	-	-	-
PC7. conduct stability studies, generate stability data & prepare stability reports for innovation products	-	-	-	-
PC8. support execution of allocated product development responsibilities	-	-	-	-
PC9. perform standardization, optimization and scale-up of API synthesis reaction	-	-	-	-
PC10. monitor reaction work-up and extraction process for API synthesis	-	-	-	-
PC11. monitor and conduct regular checks on equipment and instrument conditions, document calibrations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required	-	-	-	-
PC13. perform all the experimentation in adherence to safety, GRP and GLP guidelines	-	-	-	-
PC14. identify issues on projects and report to higher authorities	-	-	-	-
PC15. assist in resolution of issues supported by good data analysis	-	-	-	-
NOS Total	40	20	20	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0516
NOS Name	Assist in development of new Active Pharmaceutical Ingredient (API) / Synthetic Product Development
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical and Contract Research, Biotechnology
Occupation	Research and Development
NSQF Level	5.5
Credits	3.0
Version	2.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0573: Assist in Medicinal Chemistry research

Description

This Job function is about the job holder assisting in medicinal chemistry research.

Scope

The scope covers the following :

- The Scope Covers the following:
- Pre product development activities
- New medicinal product development activities

Elements and Performance Criteria

Pre product development activities

To be competent, the user/individual on the job must be able to:

- PC1.** support in developing exploratory development proposals for new innovation products
- PC2.** perform a literature search for the new product development
- PC3.** support in carrying out technical feasibility analysis on the NPD ideas
- PC4.** collect and analyze current development plans, new scientific developments and competitive activities
- PC5.** carryout research on assigned research problems contributing to the attainment of a defined research objective

New medicinal product development activities

To be competent, the user/individual on the job must be able to:

- PC6.** support in execution of product (innovation) development related activities
- PC7.** conduct stability studies, generate stability data & prepare stability reports for innovation products
- PC8.** support execution of allocated product development responsibilities
- PC9.** perform standardization, optimization and scale-up of medicinal product synthesis reaction
- PC10.** monitor reaction work-up and extraction process for new medicinal product synthesis
- PC11.** monitor and conduct regular checks on equipment and instrument conditions, document calibrations
- PC12.** use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required
- PC13.** perform all the experimentation in adherence to safety, GRP and GLP guidelines
- PC14.** identify issues on projects and report to higher authorities
- PC15.** assist in resolution of issues supported by good data analysis

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** Understanding of the company's strategic goals and product development objectives.
- KU2.** how to to extract relevant information from scientific literature.
- KU3.** Knowledge of product development methodologies and feasibility assessment.
- KU4.** Market analysis and competitive intelligence knowledge.
- KU5.** understanding of Good Research Practice (GRP) and Good Laboratory Practice (GLP) standards.
- KU6.** Familiarity with safety procedures related to equipment operation.
- KU7.** Awareness of personal protective equipment (PPE) requirements for different tasks.
- KU8.** Knowledge of chemical processes and reaction monitoring.
- KU9.** Knowledge of Material Safety Data Sheets (MSDS) information

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre product development activities</i>	20	20	5	5
PC1. support in developing exploratory development proposals for new innovation products	-	-	-	-
PC2. perform a literature search for the new product development	-	-	-	-
PC3. support in carrying out technical feasibility analysis on the NPD ideas	-	-	-	-
PC4. collect and analyze current development plans, new scientific developments and competitive activities	-	-	-	-
PC5. carryout research on assigned research problems contributing to the attainment of a defined research objective	-	-	-	-
<i>New medicinal product development activities</i>	20	20	5	5
PC6. support in execution of product (innovation) development related activities	-	-	-	-
PC7. conduct stability studies, generate stability data & prepare stability reports for innovation products	-	-	-	-
PC8. support execution of allocated product development responsibilities	-	-	-	-
PC9. perform standardization, optimization and scale-up of medicinal product synthesis reaction	-	-	-	-
PC10. monitor reaction work-up and extraction process for new medicinal product synthesis	-	-	-	-
PC11. monitor and conduct regular checks on equipment and instrument conditions, document calibrations	-	-	-	-
PC12. use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. perform all the experimentation in adherence to safety, GRP and GLP guidelines	-	-	-	-
PC14. identify issues on projects and report to higher authorities	-	-	-	-
PC15. assist in resolution of issues supported by good data analysis	-	-	-	-
NOS Total	40	40	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0573
NOS Name	Assist in Medicinal Chemistry research
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical
Occupation	Research and Development
NSQF Level	5.5
Credits	2.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0574: Assist in new Non Sterile formulation product development

Description

This Job function is about the job holder assisting in new Non Sterile formulation product development

Scope

The scope covers the following :

- The Scope Covers the following :
- Pre product development activities
- New Non-Sterile formulations development activities

Elements and Performance Criteria

Pre product development activities

To be competent, the user/individual on the job must be able to:

- PC1.** support in developing exploratory development proposals for new innovation products
- PC2.** perform a literature search for the new Nonsterile product development
- PC3.** support in carrying out technical feasibility analysis on the NPD ideas
- PC4.** collect and analyze current development plans, new scientific developments and competitive activities
- PC5.** carryout research on assigned research problems contributing to the attainment of a defined research objective

New Non-Sterile formulations development activities

To be competent, the user/individual on the job must be able to:

- PC6.** support in execution of Nonsterile product (innovation) development related activities
- PC7.** conduct stability studies, generate stability data & prepare stability reports for innovation products
- PC8.** support execution of allocated Nonsterile product development responsibilities
- PC9.** perform standardization, optimization and scale-up of new Nonsterile formulation synthesis reaction
- PC10.** monitor reaction work-up and extraction process for new Nonsterile formulation synthesis
- PC11.** monitor and conduct regular checks on equipment and instrument conditions, document calibrations
- PC12.** use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required
- PC13.** perform all the experimentation in adherence to safety, GRP and GLP guidelines
- PC14.** identify issues on projects and report to higher authorities
- PC15.** assist in resolution of issues supported by good data analysis

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Understanding of market research and customer needs analysis.
- KU2.** Awareness of regulatory and compliance requirements for non-sterile product development.
- KU3.** Knowledge of industry-specific factors related to non-sterile products.
- KU4.** Familiarity with data collection and analysis techniques.
- KU5.** Understanding of the non-sterile product development process
- KU6.** Knowledge of stability testing protocols for non-sterile products.
- KU7.** Knowledge of chemical synthesis principles for non-sterile formulations.
- KU8.** Ability to troubleshoot and problem-solve in formulation synthesis.
- KU9.** Familiarity with safety procedures related to equipment operation.
- KU10.** Understanding of personal protective equipment (PPE) requirements for different tasks.
- KU11.** Understanding of Good Research Practice (GRP) and Good Laboratory Practice (GLP) standards.
- KU12.** understanding of handling hazardous materials in a non-sterile environment.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre product development activities</i>	20	20	5	5
PC1. support in developing exploratory development proposals for new innovation products	-	-	-	-
PC2. perform a literature search for the new Nonsterile product development	-	-	-	-
PC3. support in carrying out technical feasibility analysis on the NPD ideas	-	-	-	-
PC4. collect and analyze current development plans, new scientific developments and competitive activities	-	-	-	-
PC5. carryout research on assigned research problems contributing to the attainment of a defined research objective	-	-	-	-
<i>New Non-Sterile formulations development activities</i>	20	20	5	5
PC6. support in execution of Nonsterile product (innovation) development related activities	-	-	-	-
PC7. conduct stability studies, generate stability data & prepare stability reports for innovation products	-	-	-	-
PC8. support execution of allocated Nonsterile product development responsibilities	-	-	-	-
PC9. perform standardization, optimization and scale-up of new Nonsterile formulation synthesis reaction	-	-	-	-
PC10. monitor reaction work-up and extraction process for new Nonsterile formulation synthesis	-	-	-	-
PC11. monitor and conduct regular checks on equipment and instrument conditions, document calibrations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required	-	-	-	-
PC13. perform all the experimentation in adherence to safety, GRP and GLP guidelines	-	-	-	-
PC14. identify issues on projects and report to higher authorities	-	-	-	-
PC15. assist in resolution of issues supported by good data analysis	-	-	-	-
NOS Total	40	40	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0574
NOS Name	Assist in new Non Sterile formulation product development
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical
Occupation	Research and Development
NSQF Level	5.5
Credits	3.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0575: Assist in aseptic product development

Description

This job function is about the job holder assisting in aseptic product development

Scope

The scope covers the following :

- The Scope Covers the following :
- Pre product development activities
- New aseptic formulations development activities

Elements and Performance Criteria

Pre product development activities

To be competent, the user/individual on the job must be able to:

- PC1.** support in developing exploratory development proposals for new innovation products
- PC2.** perform a literature search for the new Aseptic product development
- PC3.** support in carrying out technical feasibility analysis on the NPD ideas
- PC4.** collect and analyze current development plans, new scientific developments and competitive activities
- PC5.** carryout research on assigned research problems contributing to the attainment of a defined research objective

New aseptic formulations development activities

To be competent, the user/individual on the job must be able to:

- PC6.** support in execution of Aseptic product (innovation) development related activities
- PC7.** conduct stability studies, generate stability data & prepare stability reports for innovation products
- PC8.** support execution of allocated Aseptic product development responsibilities
- PC9.** perform standardization, optimization and scale-up of new Aseptic product synthesis reaction
- PC10.** monitor reaction work-up and extraction process for new Aseptic product synthesis
- PC11.** monitor and conduct regular checks on equipment and instrument conditions, document calibrations
- PC12.** use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required
- PC13.** perform all the experimentation in adherence to safety, GRP and GLP guidelines
- PC14.** identify issues on projects and report to higher authorities
- PC15.** assist in resolution of issues supported by good data analysis

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Knowledge of the product development process, including idea generation and proposal development.
- KU2.** Research skills, including using databases and search engines to find relevant scientific literature.
- KU3.** Knowledge of aseptic product development principles and terminology.
- KU4.** Knowledge of the factors that determine the feasibility of a product development idea.
- KU5.** Ability to identify relevant scientific and technological advancements.
- KU6.** Understanding of aseptic product development processes as per SOPs
- KU7.** Knowledge of stability testing protocols.
- KU8.** Knowledge of chemical processes and synthesis.
- KU9.** Familiarity with laboratory and production-scale equipment.
- KU10.** Knowledge of laboratory equipment and instrumentation.
- KU11.** Knowledge of safety regulations, including Good Laboratory Practices (GLP) and Good Research Practices (GRP).
- KU12.** Understanding of root cause analysis and corrective action

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre product development activities</i>	20	20	5	5
PC1. support in developing exploratory development proposals for new innovation products	-	-	-	-
PC2. perform a literature search for the new Aseptic product development	-	-	-	-
PC3. support in carrying out technical feasibility analysis on the NPD ideas	-	-	-	-
PC4. collect and analyze current development plans, new scientific developments and competitive activities	-	-	-	-
PC5. carryout research on assigned research problems contributing to the attainment of a defined research objective	-	-	-	-
<i>New aseptic formulations development activities</i>	20	20	5	5
PC6. support in execution of Aseptic product (innovation) development related activities	-	-	-	-
PC7. conduct stability studies, generate stability data & prepare stability reports for innovation products	-	-	-	-
PC8. support execution of allocated Aseptic product development responsibilities	-	-	-	-
PC9. perform standardization, optimization and scale-up of new Aseptic product synthesis reaction	-	-	-	-
PC10. monitor reaction work-up and extraction process for new Aseptic product synthesis	-	-	-	-
PC11. monitor and conduct regular checks on equipment and instrument conditions, document calibrations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required	-	-	-	-
PC13. perform all the experimentation in adherence to safety, GRP and GLP guidelines	-	-	-	-
PC14. identify issues on projects and report to higher authorities	-	-	-	-
PC15. assist in resolution of issues supported by good data analysis	-	-	-	-
NOS Total	40	40	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0575
NOS Name	Assist in aseptic product development
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical
Occupation	Research and Development
NSQF Level	5.5
Credits	2.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0576: Assist in new biological product development

Description

This job function is about the job holder assisting in new biological product development

Scope

The scope covers the following :

- The Scope Covers the following:
- Pre product development activities
- New biological product development activities

Elements and Performance Criteria

Pre product development activities

To be competent, the user/individual on the job must be able to:

- PC1.** support in developing exploratory development proposals for new innovation products
- PC2.** perform a literature search for the new product development
- PC3.** support in carrying out technical feasibility analysis on the NPD ideas
- PC4.** collect and analyze current development plans, new scientific developments and competitive activities
- PC5.** carryout research on assigned research problems contributing to the attainment of a defined research objective

New biological product development activities

To be competent, the user/individual on the job must be able to:

- PC6.** support in execution of product (innovation) development related activities
- PC7.** conduct stability studies, generate stability data & prepare stability reports for innovation products
- PC8.** support execution of allocated product development responsibilities
- PC9.** perform standardization ,optimization and scale-up of biological product synthesis reaction
- PC10.** monitor reaction work-up and extraction process for new biological product synthesis
- PC11.** monitor and conduct regular checks on equipment and instrument conditions, document calibrations
- PC12.** use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required
- PC13.** perform all the experimentation in adherence to safety ,GRP and GLP guidelines
- PC14.** identify issues on projects and report to higher authorities
- PC15.** assist in resolution of issues supported by good data analysis

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Understanding of the product development lifecycle and innovation processes.
- KU2.** Research and information retrieval skills using scientific databases and resources.
- KU3.** Familiarity with biological product development terminology and relevant literature.
- KU4.** Skills to assess technical, regulatory, and market feasibility.
- KU5.** Research methodology, experiment design, and data collection skills.
- KU6.** Knowledge of laboratory and pilot-scale equipment and procedures as per SOPs
- KU7.** Knowledge of stability testing protocols, including ICH guidelines.
- KU8.** Ability to work collaboratively in a research and development environment.
- KU9.** Knowledge of chemical engineering processes.
- KU10.** Knowledge of equipment maintenance and calibration procedures.
- KU11.** Knowledge of safety regulations, including Good Laboratory Practices (GLP) and Good Research Practices (GRP).
- KU12.** Understanding of root cause analysis and data-driven decision-making.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre product development activities</i>	20	20	5	5
PC1. support in developing exploratory development proposals for new innovation products	-	-	-	-
PC2. perform a literature search for the new product development	-	-	-	-
PC3. support in carrying out technical feasibility analysis on the NPD ideas	-	-	-	-
PC4. collect and analyze current development plans, new scientific developments and competitive activities	-	-	-	-
PC5. carryout research on assigned research problems contributing to the attainment of a defined research objective	-	-	-	-
<i>New biological product development activities</i>	20	20	5	5
PC6. support in execution of product (innovation) development related activities	-	-	-	-
PC7. conduct stability studies, generate stability data & prepare stability reports for innovation products	-	-	-	-
PC8. support execution of allocated product development responsibilities	-	-	-	-
PC9. perform standardization ,optimization and scale-up of biological product synthesis reaction	-	-	-	-
PC10. monitor reaction work-up and extraction process for new biological product synthesis	-	-	-	-
PC11. monitor and conduct regular checks on equipment and instrument conditions, document calibrations	-	-	-	-
PC12. use appropriate measuring instruments, equipment, tools, accessories, MSDS and relevant personal protective equipment (PPE) as required	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. perform all the experimentation in adherence to safety ,GRP and GLP guidelines	-	-	-	-
PC14. identify issues on projects and report to higher authorities	-	-	-	-
PC15. assist in resolution of issues supported by good data analysis	-	-	-	-
NOS Total	40	40	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0576
NOS Name	Assist in new biological product development
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Biotechnology
Occupation	Research and Development
NSQF Level	5.5
Credits	5.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0517: Assist in technology translation and process development

Description

This NOS is about a job holder assisting in technology translation and process development

Scope

The scope covers the following :

- The Scope Covers the following :
- Initiation and planning activities
- Technology Transfer activities

Elements and Performance Criteria

Initiation and planning activities

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the need for the technology transfer
- PC2.** perform a risk assessment based on the available data, information and knowledge of the premises, materials, products, procedures and other related information
- PC3.** prepare the technology transfer document such as the project plan
- PC4.** assist in development of a control strategy which includes, risks; raw, starting and packaging material attributes; analytical procedures in QC; critical quality attributes (CQAs), critical process parameters (CPPs) and in-process controls; and acceptance criteria and limits
- PC5.** ensure Drug Master File (DMF), API Master File (APIMF), or equivalent information, as well as any relevant additional information importance for the manufacture of the pharmaceutical product available to team
- PC6.** ensure the critical quality attributes, critical process parameters, material attributes, control strategy and any other impacting elements on the quality of the product should be available before the transfer process

Technology Transfer activities

To be competent, the user/individual on the job must be able to:

- PC7.** devise ways to produce the new product on a large scale with standardized protocols and support in technology transfer
- PC8.** research and develop ways to manufacture products and monitor existing processes and products for quality and efficiency
- PC9.** assess the manufacturer's information like raw materials, critical process parameters, performing equipment and comparisons
- PC10.** review packaging line trials, e.g. if a product is a new SKU to the plant/site it is crucial to perform line trials for filling the product into the proposed bottle/ container
- PC11.** review stability indicating methods to enable onsite transfer against the protocol signed off by the client/manufacturer/ R&D team
- PC12.** review validation records for the small scale batch produced
- PC13.** crosscheck safety handling protocols for the product



Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of pharmaceutical product development and commercialization processes.
- KU2.** Familiarity with pharmaceutical manufacturing and supply chain risks.
- KU3.** Understanding of pharmaceutical product attributes and quality standards.
- KU4.** Familiarity with DMF and APIMF requirements.
- KU5.** Understanding of manufacturing processes, equipment, and raw materials.
- KU6.** ability to ensure the compatibility of products with packaging.
- KU7.** Knowledge of stability testing protocols.
- KU8.** Understanding of the importance of validation in product transfer.
- KU9.** Knowledge of safety and handling protocols for pharmaceutical products

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Initiation and planning activities</i>	20	10	10	10
PC1. Identify the need for the technology transfer	-	-	-	-
PC2. perform a risk assessment based on the available data, information and knowledge of the premises, materials, products, procedures and other related information	-	-	-	-
PC3. prepare the technology transfer document such as the project plan	-	-	-	-
PC4. assist in development of a control strategy which includes, risks; aw, starting and packaging material attributes; analytical procedures in QC; critical quality attributes (CQAs), critical process parameters (CPPs) and in-process controls; and acceptance criteria and limits	-	-	-	-
PC5. ensure Drug Master File (DMF), API Master File (APIMF), or equivalent information, as well as any relevant additional information importance for the manufacture of the pharmaceutical product available to team	-	-	-	-
PC6. ensure the critical quality attributes, critical process parameters, material attributes, control strategy and any other impacting elements on the quality of the product should be available before the transfer process	-	-	-	-
<i>Technology Transfer activities</i>	20	10	10	10
PC7. devise ways to produce the new product on a large scale with standardized protocols and support in technology transfer	-	-	-	-
PC8. research and develop ways to manufacture products and monitor existing processes and products for quality and efficiency	-	-	-	-
PC9. asses the manufacturer's information like raw materials, critical process parameters, performing equipment and comparisons	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. review packaging line trials, e.g. if a product is a new SKU to the plant/site it is crucial to perform line trials for filling the product into the proposed bottle/ container	-	-	-	-
PC11. review stability indicating methods to enable onsite transfer against the protocol signed off by the client/manufacturer/ R&D team	-	-	-	-
PC12. review validation records for the small scale batch produced	-	-	-	-
PC13. crosscheck safety handling protocols for the product	-	-	-	-
NOS Total	40	20	20	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0517
NOS Name	Assist in technology translation and process development
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Biotechnology, Bio Pharmaceutical and Contract Research
Occupation	Research and Development
NSQF Level	5.5
Credits	5.0
Version	2.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0577: Assist in technology transfer

Description

This job function is about the job holder assisting in technology transfer

Scope

The scope covers the following :

- The Scope Covers the following:
- Initiation and planning activities
- Technology Transfer activities

Elements and Performance Criteria

Initiation and planning activities

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the need for the technology transfer
- PC2.** perform a risk assessment based on the available data, information and knowledge of the premises, materials, products, procedures and other related information.
- PC3.** prepare the technology transfer document such as the project plan
- PC4.** assist in development of a control strategy which includes, risks; raw, starting and packaging material attributes; analytical procedures in QC; critical quality attributes (CQAs), critical process parameters (CPPs) and in-process controls; and acceptance criteria and limits
- PC5.** ensure Drug Master File (DMF), API Master File (APIMF), or equivalent information, as well as any relevant additional information importance for the manufacture of the pharmaceutical product available to team
- PC6.** ensure the critical quality attributes, critical process parameters, material attributes, control strategy and any other impacting elements on the quality of the product should be available before the transfer process

Technology Transfer activities

To be competent, the user/individual on the job must be able to:

- PC7.** devise ways to produce the new product on a large scale with standardized protocols and support in technology transfer
- PC8.** research and develop ways to manufacture products and monitor existing processes and products for quality and efficiency
- PC9.** assess the manufacturer's information like raw materials, critical process parameters, performing equipment and comparisons
- PC10.** review packaging line trials, e.g. if a product is a new SKU to the plant/site it is crucial to perform line trials for filling the product into the proposed bottle/ container
- PC11.** review stability indicating methods to enable onsite transfer against the protocol signed off by the client/manufacturer/ R&D team
- PC12.** Preview validation records for the small scale batch produced
- PC13.** crosscheck safety handling protocols for the product



Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of pharmaceutical product development and commercialization processes
- KU2.** Familiarity with pharmaceutical manufacturing and supply chain risks.
- KU3.** Understanding of pharmaceutical product attributes and quality standards.
- KU4.** Familiarity with DMF and APIMF requirements.
- KU5.** Understanding of manufacturing processes, equipment, and raw materials.
- KU6.** ability to ensure the compatibility of products with packaging.
- KU7.** Knowledge of stability testing protocols.
- KU8.** Understanding of the importance of validation in product transfer
- KU9.** Knowledge of safety and handling protocols for pharmaceutical products

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Initiation and planning activities</i>	20	10	10	10
PC1. Identify the need for the technology transfer	-	-	-	-
PC2. perform a risk assessment based on the available data, information and knowledge of the premises, materials, products, procedures and other related information.	-	-	-	-
PC3. prepare the technology transfer document such as the project plan	-	-	-	-
PC4. assist in development of a control strategy which includes, risks; aw, starting and packaging material attributes; analytical procedures in QC; critical quality attributes (CQAs), critical process parameters (CPPs) and in-process controls; and acceptance criteria and limits	-	-	-	-
PC5. ensure Drug Master File (DMF), API Master File (APIMF), or equivalent information, as well as any relevant additional information importance for the manufacture of the pharmaceutical product available to team	-	-	-	-
PC6. ensure the critical quality attributes, critical process parameters, material attributes, control strategy and any other impacting elements on the quality of the product should be available before the transfer process	-	-	-	-
<i>Technology Transfer activities</i>	20	10	10	10
PC7. devise ways to produce the new product on a large scale with standardized protocols and support in technology transfer	-	-	-	-
PC8. research and develop ways to manufacture products and monitor existing processes and products for quality and efficiency	-	-	-	-
PC9. asses the manufacturer's information like raw materials, critical process parameters, performing equipment and comparisons	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. review packaging line trials, e.g. if a product is a new SKU to the plant/site it is crucial to perform line trials for filling the product into the proposed bottle/ container	-	-	-	-
PC11. review stability indicating methods to enable onsite transfer against the protocol signed off by the client/manufacturer/ R&D team	-	-	-	-
PC12. Preview validation records for the small scale batch produced	-	-	-	-
PC13. crosscheck safety handling protocols for the product	-	-	-	-
NOS Total	40	20	20	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0577
NOS Name	Assist in technology transfer
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Biotechnology
Occupation	Research and Development
NSQF Level	5.5
Credits	5.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0578: Conducts research and analysis using mass spectrometry techniques to solve complex problems, develop new products, or improve existing ones.

Description

This job function is about the job holder involves in Conducting research and analysis using mass spectrometry techniques to solve complex problems, develop new products, or improve existing ones.

Scope

The scope covers the following :

- The Scope Covers the following:
- Mass Spectrometry Analysis

Elements and Performance Criteria

Mass Spectrometry Analysis

To be competent, the user/individual on the job must be able to:

- PC1.** prepare the samples for analysis by Mass Spectrometry
- PC2.** select the correct developed and validated method for analysis
- PC3.** Operate the UV spectrophotometer as per SOP
- PC4.** handle cuvette with loaded samples
- PC5.** record the readings for different samples and make the entry in instrument logbook
- PC6.** Ensure that mass spectrometry data is consistently accurate, reliable, and precise.
- PC7.** Maintain the highest standards of data integrity and quality throughout the analytical process.
- PC8.** Possess expertise in designing and improving analytical methods to achieve higher efficiency and effectiveness. Continuously seek ways to enhance method sensitivity, selectivity, and robustness.
- PC9.** Demonstrate mastery in the operation, maintenance, and troubleshooting of mass spectrometry instruments. Ensure that equipment is calibrated and functioning optimally for accurate results
- PC10.** Adhere to regulatory guidelines (e.g., FDA, EPA) in all analytical work. Keep thorough and well-organized records to meet compliance requirements and facilitate audits.
- PC11.** Contribute valuable insights and expertise to research projects.
- PC12.** Collaborate effectively with cross-functional teams to advance project objectives, ensuring that mass spectrometry data supports research goals.
- PC13.** Properly handle and dispose of hazardous materials in compliance with regulations, all while minimizing the environmental impact of laboratory practices.
- PC14.** Actively participate in conferences, workshops, and training sessions to stay abreast of industry advancements and emerging technologies

Qualification Pack

- PC15.** communicate Clearly and comprehensively mass spectrometry findings to team members, supervisors, and stakeholders.
- PC16.** Generate reports that include detailed data, methods, and interpretations. Actively contribute insights to facilitate informed decision-making processes.
- PC17.** Demonstrate a strong ability to identify and resolve technical challenges swiftly and effectively.
- PC18.** Implement innovative solutions to analytical issues, minimizing any potential downtime in laboratory operations.
- PC19.** Safeguard confidential information and proprietary data while maintaining the utmost integrity in every professional interaction

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of sample preparation techniques, including extraction and purification.
- KU2.** Familiarity with various analytical methods and their applicability.
- KU3.** Knowledge of instrument logbook maintenance.
- KU4.** Knowledge of mass spectrometry principles and instrumentation.
- KU5.** Understanding of data integrity principles and regulatory requirements.
- KU6.** Knowledge of hazardous materials and waste disposal regulations
- KU7.** Understanding of UV spectrophotometer principles and operation.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Mass Spectrometry Analysis</i>	40	40	10	10
PC1. prepare the samples for analysis by Mass Spectrometry	-	-	-	-
PC2. select the correct developed and validated method for analysis	-	-	-	-
PC3. Operate the UV spectrophotometer as per SOP	-	-	-	-
PC4. handle cuvette with loaded samples	-	-	-	-
PC5. record the readings for different samples and make the entry in instrument logbook	-	-	-	-
PC6. Ensure that mass spectrometry data is consistently accurate, reliable, and precise.	-	-	-	-
PC7. Maintain the highest standards of data integrity and quality throughout the analytical process.	-	-	-	-
PC8. Possess expertise in designing and improving analytical methods to achieve higher efficiency and effectiveness. Continuously seek ways to enhance method sensitivity, selectivity, and robustness.	-	-	-	-
PC9. Demonstrate mastery in the operation, maintenance, and troubleshooting of mass spectrometry instruments. Ensure that equipment is calibrated and functioning optimally for accurate results	-	-	-	-
PC10. Adhere to regulatory guidelines (e.g., FDA, EPA) in all analytical work. Keep thorough and well-organized records to meet compliance requirements and facilitate audits.	-	-	-	-
PC11. Contribute valuable insights and expertise to research projects.	-	-	-	-
PC12. Collaborate effectively with cross-functional teams to advance project objectives, ensuring that mass spectrometry data supports research goals.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Properly handle and dispose of hazardous materials in compliance with regulations, all while minimizing the environmental impact of laboratory practices.	-	-	-	-
PC14. Actively participate in conferences, workshops, and training sessions to stay abreast of industry advancements and emerging technologies	-	-	-	-
PC15. communicate Clearly and comprehensively mass spectrometry findings to team members, supervisors, and stakeholders.	-	-	-	-
PC16. Generate reports that include detailed data, methods, and interpretations. Actively contribute insights to facilitate informed decision-making processes.	-	-	-	-
PC17. Demonstrate a strong ability to identify and resolve technical challenges swiftly and effectively.	-	-	-	-
PC18. Implement innovative solutions to analytical issues, minimizing any potential downtime in laboratory operations.	-	-	-	-
PC19. Safeguard confidential information and proprietary data while maintaining the utmost integrity in every professional interaction	-	-	-	-
NOS Total	40	40	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0578
NOS Name	Conducts research and analysis using mass spectrometry techniques to solve complex problems, develop new products, or improve existing ones.
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Biotechnology
Occupation	Research and Development
NSQF Level	5.5
Credits	5.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0579: Conducts research and analysis using Nuclear Magnetic Resonance techniques to solve complex problems, develop new products, or improve existing ones.

Description

This job function is about the job holder involved in conducting research and analysis using Nuclear Magnetic Resonance techniques to solve complex problems, develop new products, or improve existing ones.

Scope

The scope covers the following :

- The Scope Covers the following:
- Research and analysis using Nuclear Magnetic Resonance

Elements and Performance Criteria

Research and analysis using Nuclear Magnetic Resonance

To be competent, the user/individual on the job must be able to:

- PC1.** Consistently generate precise and reliable NMR spectra, ensuring data accuracy and integrity in complex analyses
- PC2.** Demonstrate proficiency in designing and optimizing NMR methods, tailoring them to specific research requirements, and enhancing method sensitivity, selectivity, and efficiency.
- PC3.** Exhibit expert-level competence in operating, maintaining, and troubleshooting NMR equipment, conducting routine calibrations, and promptly addressing instrument issues to ensure uninterrupted analyses.
- PC4.** Strictly adhere to relevant regulatory guidelines, such as FDA or EPA regulations, during NMR analyses, while maintaining meticulous records to support compliance and facilitate audits.
- PC5.** Prioritize laboratory safety protocols, minimizing risks, and adhere to environmentally responsible practices, including the proper handling and disposal of hazardous materials.
- PC6.** Actively engage in research projects, providing valuable NMR insights, and collaborate effectively with cross-functional teams, contributing to the achievement of project objectives and meeting deadlines.
- PC7.** Pursue continuous professional development by actively participating in conferences, workshops, and training sessions to stay updated on evolving NMR techniques and emerging technologies
- PC8.** Clearly and comprehensively communicate NMR findings to team members, supervisors, and stakeholders, generating detailed reports that encompass data, methods, and interpretations to facilitate informed decision-making
- PC9.** Demonstrate exceptional problem-solving skills, swiftly identifying and resolving technical challenges related to NMR analyses, and implementing innovative solutions to minimize downtime and maintain productivity



Qualification Pack

PC10. Uphold the highest ethical and professional standards in all aspects of scientific research and analysis, safeguarding confidential information and ensuring data integrity with the utmost integrity and discretion

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** In-depth knowledge of NMR principles and techniques.
- KU2.** Knowledge of parameter adjustments to improve method performance
- KU3.** Understanding and Expertise in instrument calibration and performance assessment.
- KU4.** Understanding of laboratory safety protocols and best practices.
- KU5.** Awareness of cutting-edge NMR technologies and trends
- KU6.** Compliance with data security and privacy standards, maintaining the highest level of professionalism.
- KU7.** Environmental responsibility and waste minimization practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Research and analysis using Nuclear Magnetic Resonance</i>	40	40	10	10
PC1. Consistently generate precise and reliable NMR spectra, ensuring data accuracy and integrity in complex analyses	-	-	-	-
PC2. Demonstrate proficiency in designing and optimizing NMR methods, tailoring them to specific research requirements, and enhancing method sensitivity, selectivity, and efficiency.	-	-	-	-
PC3. Exhibit expert-level competence in operating, maintaining, and troubleshooting NMR equipment, conducting routine calibrations, and promptly addressing instrument issues to ensure uninterrupted analyses.	-	-	-	-
PC4. Strictly adhere to relevant regulatory guidelines, such as FDA or EPA regulations, during NMR analyses, while maintaining meticulous records to support compliance and facilitate audits.	-	-	-	-
PC5. Prioritize laboratory safety protocols, minimizing risks, and adhere to environmentally responsible practices, including the proper handling and disposal of hazardous materials.	-	-	-	-
PC6. Actively engage in research projects, providing valuable NMR insights, and collaborate effectively with cross-functional teams, contributing to the achievement of project objectives and meeting deadlines.	-	-	-	-
PC7. Pursue continuous professional development by actively participating in conferences, workshops, and training sessions to stay updated on evolving NMR techniques and emerging technologies	-	-	-	-
PC8. Clearly and comprehensively communicate NMR findings to team members, supervisors, and stakeholders, generating detailed reports that encompass data, methods, and interpretations to facilitate informed decision-making	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Demonstrate exceptional problem-solving skills, swiftly identifying and resolving technical challenges related to NMR analyses, and implementing innovative solutions to minimize downtime and maintain productivity	-	-	-	-
PC10. Uphold the highest ethical and professional standards in all aspects of scientific research and analysis, safeguarding confidential information and ensuring data integrity with the utmost integrity and discretion	-	-	-	-
NOS Total	40	40	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0579
NOS Name	Conducts research and analysis using Nuclear Magnetic Resonance techniques to solve complex problems, develop new products, or improve existing ones.
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Biotechnology
Occupation	Research and Development
NSQF Level	5.5
Credits	5.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

LFS/N0126: Establish innovation led enterprise and perform various entrepreneurial activities to run the regulated business operations

Description

This NOS is about the entrepreneur setting up his/her own licensed innovation led enterprise and performing various entrepreneurial activities to run the regulated business operations

Scope

The scope covers the following :

- Set up innovation led enterprises and perform entrepreneurial activities
- Maintenance of accounts and ledgers
- Comply with legal, regulatory, and statutory standards

Elements and Performance Criteria

Set up innovation led enterprise and perform entrepreneurial activities

To be competent, the user/individual on the job must be able to:

- PC1.** identify the innovation domain for business activities and identify prospective customers and business opportunity
- PC2.** identify innovation led products and/ or services and their sources, that match the business opportunity
- PC3.** develop a business proposal with an objective and identified business area for registering the proposed business as an independent entity with competent authorities & incubation center
- PC4.** submit business proposals and scientific feasibility plans for securing the required funding from govt. schemes for MSMEs like MUDRA etc. as and when needed
- PC5.** present the business proposal to investors and stakeholders to gain their confidence and trust for possible funding
- PC6.** ensure setting up the technological infrastructure and resources for business as per the approved plan and investors/ shareholders agreement
- PC7.** enrol into various government schemes and programs for MSME and avail the benefits
- PC8.** promote the product/ services through various means including digital promotions in line to regulations for promotion and sale as per land of law
- PC9.** develop the supply chain and distribution network in compliance to regulations
- PC10.** maintain updated information regarding new market trends to provide service to customers and constantly upskill self and employees for futuristic advance technologies, innovations, and methodologies in innovation domain.

Maintenance of accounts and ledgers

To be competent, the user/individual on the job must be able to:

- PC11.** ensure to generate a final invoice for the services rendered/ or products sold
- PC12.** collect payment either in cash (as per RBI guidelines)/ cheque/ demand draft or through online transactions via RTGS/NEFT/UPI etc.

Qualification Pack

- PC13.** ensure accounts and ledgers are maintained for the material supplied, payables and receivables, loan ledgers, taxes collected and paid into government treasury as well for wages paid and statutory benefits like PF, ESI contribution deposited with authorities
- PC14.** ensure maintenance and reconciliation of a monthly account as well as movable and immovable assets and stock to keep track of profit/ loss as well as periodic audit as per legal requirements

Comply with legal, regulatory and statutory standards

To be competent, the user/individual on the job must be able to:

- PC15.** comply with workplace health and safety rules stipulated by local authorities
- PC16.** comply with rules related to taxes and licensing regulations
- PC17.** comply with prevalent labour laws and keep records for employee's qualifications, employment as well as trainings
- PC18.** comply with regulations and protocols related to maintenance of product/ service license including the pre conditions (if any)
- PC19.** comply with quality systems like ISO/GMP/GLP/NCS-TCP etc and any other rule mandated by appropriate regulatory authorities
- PC20.** comply with pollution norms as per land of law and ensure near zero pollution for a sustainable environment and to earn carbon credits
- PC21.** ensure that business facility and employees are always prepared and ready for any surprise inspection/ audit from clients/ regulatory/ statutory authorities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** business administration, financial and planning activities
- KU2.** data analysis techniques for continuous improvement and identifying new business opportunities
- KU3.** maintaining confidentiality for information about finances, business secrets and information shared by clients/ stakeholders
- KU4.** procedures for funding approvals, and presenting new business idea/ opportunity for license or stakeholder approvals
- KU5.** prevailing and possible environmental issues and expected quality standards
- KU6.** license, patent and copyright laws
- KU7.** applicable tax, duties and labour laws
- KU8.** fundamentals of costing, pricing and profit
- KU9.** accounting principles and use of accounting software and government software/ portals for taxes and returns
- KU10.** Good Practices (GxP), ISO and other quality systems
- KU11.** talent management and resource planning
- KU12.** Awareness of applicable engineering concepts
- KU13.** audits and regulations in Life Sciences sector applicable for chosen business area
- KU14.** market promotion, globally trending strategies and distribution network
- KU15.** concept of innovation and jugaad principle



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information to be communicated
- GS2.** fill relevant applications where required
- GS3.** write clear communications to government officials, funding partners, team members, etc
- GS4.** discuss task lists, schedules and activities with customers, peers, partners, vendors etc.
- GS5.** effectively communicate with teams
- GS6.** interact with successful local entrepreneurs, and enterprises in a similar field for best practices and expertise
- GS7.** ask questions in order to understand problems and clarify queries
- GS8.** multi-task and adapt to meet work timelines
- GS9.** apply emotional intelligence while dealing with other genders and people with disability

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Set up innovation led enterprise and perform entrepreneurial activities</i>	20	30	6	3
PC1. identify the innovation domain for business activities and identify prospective customers and business opportunity	-	-	-	-
PC2. identify innovation led products and/ or services and their sources, that match the business opportunity	-	-	-	-
PC3. develop a business proposal with an objective and identified business area for registering the proposed business as an independent entity with competent authorities & incubation center	-	-	-	-
PC4. submit business proposals and scientific feasibility plans for securing the required funding from govt. schemes for MSMEs like MUDRA etc. as and when needed	-	-	-	-
PC5. present the business proposal to investors and stakeholders to gain their confidence and trust for possible funding	-	-	-	-
PC6. ensure setting up the technological infrastructure and resources for business as per the approved plan and investors/ shareholders agreement	-	-	-	-
PC7. enrol into various government schemes and programs for MSME and avail the benefits	-	-	-	-
PC8. promote the product/ services through various means including digital promotions in line to regulations for promotion and sale as per land of law	-	-	-	-
PC9. develop the supply chain and distribution network in compliance to regulations	-	-	-	-
PC10. maintain updated information regarding new market trends to provide service to customers and constantly upskill self and employees for futuristic advance technologies, innovations, and methodologies in innovation domain.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintenance of accounts and ledgers</i>	10	20	6	5
PC11. ensure to generate a final invoice for the services rendered/ or products sold	-	-	-	-
PC12. collect payment either in cash (as per RBI guidelines)/ cheque/ demand draft or through online transactions via RTGS/NEFT/UPI etc.	-	-	-	-
PC13. ensure accounts and ledgers are maintained for the material supplied, payables and receivables, loan ledgers, taxes collected and paid into government treasury as well for wages paid and statutory benefits like PF, ESI contribution deposited with authorities	-	-	-	-
PC14. ensure maintenance and reconciliation of a monthly account as well as movable and immovable assets and stock to keep track of profit/ loss as well as periodic audit as per legal requirements	-	-	-	-
<i>Comply with legal, regulatory and statutory standards</i>	-	-	-	-
PC15. comply with workplace health and safety rules stipulated by local authorities	-	-	-	-
PC16. comply with rules related to taxes and licensing regulations	-	-	-	-
PC17. comply with prevalent labour laws and keep records for employee's qualifications, employment as well as trainings	-	-	-	-
PC18. comply with regulations and protocols related to maintenance of product/ service license including the pre conditions (if any)	-	-	-	-
PC19. comply with quality systems like ISO/GMP/GLP/NCS-TCP etc and any other rule mandated by appropriate regulatory authorities	-	-	-	-
PC20. comply with pollution norms as per land of law and ensure near zero pollution for a sustainable environment and to earn carbon credits	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. ensure that business facility and employees are always prepared and ready for any surprise inspection/ audit from clients/ regulatory/ statutory authorities	-	-	-	-
NOS Total	30	50	12	8



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0126
NOS Name	Establish innovation led enterprise and perform various entrepreneurial activities to run the regulated business operations
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Contract Research, Biotechnology
Occupation	Generic
NSQF Level	5.5
Credits	4.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023



Qualification Pack

LFS/N0127: Maintain the Critical Business documents as Innovation led entrepreneur

Description

This NOS is about an entrepreneur maintaining the critical business documents for its Innovation led business

Scope

The scope covers the following :

- Infrastructure documents
- Supply Chain documents
- Documentation for sales & marketing
- Quality audit and client/regulatory inspections related documentation
- IPR related Documents

Elements and Performance Criteria

Infrastructure documents

To be competent, the user/individual on the job must be able to:

- PC1.** ensure documentation and maintenance of records for architectural building layout like blueprint of plant/lab, electric circuit and plumbing layouts and safety equipment layout, infrastructure and asset insurance and lease/rent contracts/ sale deed (if owned facility)
- PC2.** ensure documentation and maintenance of records for all machineries, equipment and tools installed/used in the plant/ lab unit, with supplier/manufacturer details, manuals of all machineries / equipment, installation, qualification calibration and validation records and annual maintenance details etc
- PC3.** ensure documentation and maintenance of records for engineering and maintenance of each machinery/equipment, machine utilization, machine performance, breakdown details, corrective actions, spares changed, machine/equipment condition etc

Supply Chain documents

To be competent, the user/individual on the job must be able to:

- PC4.** ensure documentation and maintenance of records for all raw materials, ingredients and packaging materials handled, like name of the supplier, batch details, quantity supplied and quality of the materials supplies etc as well as per NCS-TCP
- PC5.** ensure documentation and maintenance of records for each supplier quality report on raw materials, ingredients, packaging materials, internal and external quality report on finished products, consumer and customer complaints , corrective actions, legal documents (if any) as per NCS-TCP
- PC6.** ensure documentation and maintenance of records for storage facility, like condition of storage facility, storage parameters if any like temperature, humidity, pressure (as applicable), space utilised, stacking procedure etc

Qualification Pack

- PC7.** ensure documentation and maintenance of records for distribution details & cold chain Management like transport details, quality, hygiene and cleanliness of vehicle, quantity loaded in the vehicle, Storage condition in distribution routes, outlet details, customer/consumer details, distribution quantity, quantity returned etc.

Documentation for sales & marketing

To be competent, the user/individual on the job must be able to:

- PC8.** ensure documentation and maintenance of records for long term and short term business plans, feasibility reports, investor/shareholder agreements and disbursement and payment records for loan/grant/equity etc.
- PC9.** ensure documentation and maintenance of records for sale like customer details, customer type, location, quantity purchased by each stockist/consumer, frequency of purchase, sale details like quantity of products/brand sold in each territory

Quality audit and client/regulatory inspections related documentation

To be competent, the user/individual on the job must be able to:

- PC10.** ensure documentation and maintenance of records for internal/ external audits with its observations, notices/ feedback, action reports and final submission of audit action reports
- PC11.** ensure documentation and maintenance of records for every communication sent to or received from regulatory/ licensing authorities

IPR Related Documents

To be competent, the user/individual on the job must be able to:

- PC12.** interact with a global clientele and draft high-quality business / IP research documents, making impactful inferences after analyzing available data, and articulating them in a clear and cogent manner
- PC13.** Draft and Review IP related document that are associated with company patent registration are fulfilled accordingly to regulatory requirements
- PC14.** ensure accurate and comprehensive IPR documentation, such as patent applications, trade secret agreements, technology transfer documents, and regulatory filings are prepared and submitted
- PC15.** Ensure submission and response associated with IPR documentation, ensuring that filings, submissions, and responses are submitted within the specified timeframes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** system of documentation as per ISO/ good documentation practices and method of implementation
- KU2.** scoring, grading and accreditation system of affiliating bodies and clients
- KU3.** guidelines for facing audits and best practices for making organization audit ready
- KU4.** software and latest information technology tools for documentation and record maintenance
- KU5.** statistical tools for analysis and monitoring
- KU6.** various recording and documentation formats applicable in sales, marketing, supply chain etc
- KU7.** basic awareness of engineering drawing and architectural layouts
- KU8.** best practices in engineering and maintenance in sector



Qualification Pack

- KU9.** accounting standards and regulations
- KU10.** primary research and secondary research, focused on intellectual property landscape associated biotechnology sector
- KU11.** basics of patent / prior art searching
- KU12.** biological sequence searching (BLAST).
- KU13.** Formulate Search Strings for Identified Invention and Perform Patentability, FTO, Landscape and Invalidity Search

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record and communicate details of work done to appropriate people using written/typed report
- GS2.** interact effectively with team members
- GS3.** communicate with team members for data/ information/support/ advice needed
- GS4.** communicate with people in a form and manner and using language that is open and respectful
- GS5.** apply balanced judgement to different situations
- GS6.** plan and organize assigned work in order to drive collaboration with team members
- GS7.** effectively interact with the various stakeholders to complete assigned tasks
- GS8.** resolve any difficulties in relationships with colleagues , or get help from an appropriate person, in a way that preserves goodwill and trust
- GS9.** use team-building skills during the interaction with teammates while managing the difficult/stressful or emotional situations at work
- GS10.** apply emotional intelligence while dealing with other genders and people with disability

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Infrastructure documents</i>	5	20	5	2
PC1. ensure documentation and maintenance of records for architectural building layout like blueprint of plant/lab, electric circuit and plumbing layouts and safety equipment layout, infrastructure and asset insurance and lease/rent contracts/ sale deed (if owned facility)	-	-	-	-
PC2. ensure documentation and maintenance of records for all machineries, equipment and tools installed/used in the plant/ lab unit, with supplier/manufacturer details, manuals of all machineries / equipment, installation, qualification calibration and validation records and annual maintenance details etc	-	-	-	-
PC3. ensure documentation and maintenance of records for engineering and maintenance of each machinery/equipment, machine utilization, machine performance, breakdown details, corrective actions, spares changed, machine/equipment condition etc	-	-	-	-
<i>Supply Chain documents</i>	5	10	5	3
PC4. ensure documentation and maintenance of records for all raw materials, ingredients and packaging materials handled, like name of the supplier, batch details, quantity supplied and quality of the materials supplies etc as well as per NCS-TCP	-	-	-	-
PC5. ensure documentation and maintenance of records for each supplier quality report on raw materials, ingredients, packaging materials, internal and external quality report on finished products, consumer and customer complaints , corrective actions, legal documents (if any) as per NCS-TCP	-	-	-	-
PC6. ensure documentation and maintenance of records for storage facility, like condition of storage facility, storage parameters if any like temperature, humidity, pressure (as applicable), space utilised, stacking procedure etc	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC7. ensure documentation and maintenance of records for distribution details & cold chain Management like transport details, quality, hygiene and cleanliness of vehicle, quantity loaded in the vehicle, Storage condition in distribution routes , outlet details, customer/ consumer details, distribution quantity, quantity returned etc.	-	-	-	-
<i>Documentation for sales & marketing</i>	5	10	3	3
PC8. ensure documentation and maintenance of records for long term and short term business plans, feasibility reports, investor/shareholder agreements and disbursement and payment records for loan/grant/equity etc.	-	-	-	-
PC9. ensure documentation and maintenance of records for sale like customer details, customer type, location, quantity purchased by each stockist/consumer, frequency of purchase, sale details like quantity of products/brand sold in each territory	-	-	-	-
<i>Quality audit and client/regulatory inspections related documentation</i>	5	5	2	2
PC10. ensure documentation and maintenance of records for internal/ external audits with its observations, notices/ feedback, action reports and final submission of audit action reports	-	-	-	-
PC11. ensure documentation and maintenance of records for every communication sent to or received from regulatory/ licensing authorities	-	-	-	-
<i>IPR Related Documents</i>	3	3	2	2
PC12. interact with a global clientele and draft high-quality business / IP research documents, making impactful inferences after analyzing available data, and articulating them in a clear and cogent manner	-	-	-	-
PC13. Draft and Review IP related document that are associated with company patent registration are fulfilled accordingly to regulatory requirements	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. ensure accurate and comprehensive IPR documentation, such as patent applications, trade secret agreements, technology transfer documents, and regulatory filings are prepared and submitted	-	-	-	-
PC15. Ensure submission and response associated with IPR documentation, ensuring that filings, submissions, and responses are submitted within the specified timeframes	-	-	-	-
NOS Total	23	48	17	12



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0127
NOS Name	Maintain the Critical Business documents as Innovation led entrepreneur
Sector	Life Sciences
Sub-Sector	Biotechnology, Pharmaceutical, Bio Pharmaceutical, Contract Research
Occupation	Generic
NSQF Level	5.5
Credits	1.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023



Qualification Pack

LFS/N0128: Product commercialization and Intellectual Property Rights Management

Description

This job function is about the job holder responsible for product commercialization and Intellectual Property Rights Management

Scope

The scope covers the following :

- The Scope Covers the following:
- Product Commercialization

Elements and Performance Criteria

Product Commercialization

To be competent, the user/individual on the job must be able to:

- PC1.** Successfully bring new products to market, meeting, or exceeding revenue targets
- PC2.** Develop a comprehensive IP strategy to protect patents, trademarks, copyrights, and manage the IP portfolio strategically
- PC3.** Collaborate effectively with internal teams (R&D, legal, marketing, sales) and external partners (law firms, investors).
- PC4.** Efficiently allocate and manage budgets for product development and IP protection, identifying cost-saving opportunities.
- PC5.** Conduct thorough market research to understand customer needs and competitive landscapes, using data for informed decisions.
- PC6.** Ensure strict adherence to relevant IP laws and regulations, staying updated on legal changes.
- PC7.** Develop and track innovation related KPIs (patents filed, time to market, market share) and report progress to stakeholders.
- PC8.** Proactively identify and mitigate risks and challenges in product development and IP management.
- PC9.** Provide leadership and mentorship, conducting performance evaluations, and offering coaching for team development.
- PC10.** Prioritize stakeholder satisfaction by ensuring product quality and IP protection while maintaining strong, collaborative relationships.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of product commercialization processes.
- KU2.** Understanding of intellectual property (IP) laws and regulations.



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- KU3.** Understanding of Key Performance Indicator (KPI) development and tracking.
- KU4.** Knowledge of Risk management and mitigation strategies.
- KU5.** Understanding how to lead or collaborate within teams, including conflict resolution and motivation.
- KU6.** Understand ethical principles and apply them to decision-making, ensuring responsible and ethical behavior.
- KU7.** Knowledge of how to Develop leadership qualities, including the ability to inspire, guide, and motivate others towards common goals.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to understand the various coding systems and to read instructions, guidelines, procedures, rules, and signages to understand the procedure to be followed
- GS2.** use listening skills to follow the instructions and procedures during emergency alarms
- GS3.** use written communication skills to accurately record every information required to be reported as per SOP and GMP guidelines in the prescribed language
- GS4.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS5.** apply decision-making skills to make balanced judgments within the authority while dealing with hazards and breaches
- GS6.** use critical thinking skills to take relevant actions on the accidents and breach in compliance with EHS protocols
- GS7.** use problem-solving skills to determine the root cause and devise a corrective action plan to address it effectively.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Product Commercialization</i>	40	40	10	10
PC1. Successfully bring new products to market, meeting, or exceeding revenue targets	-	-	-	-
PC2. Develop a comprehensive IP strategy to protect patents, trademarks, copyrights, and manage the IP portfolio strategically	-	-	-	-
PC3. Collaborate effectively with internal teams (R&D, legal, marketing, sales) and external partners (law firms, investors).	-	-	-	-
PC4. Efficiently allocate and manage budgets for product development and IP protection, identifying cost-saving opportunities.	-	-	-	-
PC5. Conduct thorough market research to understand customer needs and competitive landscapes, using data for informed decisions.	-	-	-	-
PC6. Ensure strict adherence to relevant IP laws and regulations, staying updated on legal changes.	-	-	-	-
PC7. Develop and track innovation related KPIs (patents filed, time to market, market share) and report progress to stakeholders.	-	-	-	-
PC8. Proactively identify and mitigate risks and challenges in product development and IP management.	-	-	-	-
PC9. Provide leadership and mentorship, conducting performance evaluations, and offering coaching for team development.	-	-	-	-
PC10. Prioritize stakeholder satisfaction by ensuring product quality and IP protection while maintaining strong, collaborative relationships.	-	-	-	-
NOS Total	40	40	10	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0128
NOS Name	Product commercialization and Intellectual Property Rights Management
Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Contract Research, Biotechnology
Occupation	Generic
NSQF Level	5.5
Credits	4.0
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by Life Sciences Sector Skill Development Council (LSSSDC)
2. Each Element will be assigned marks proportional to its importance in NOS. LSSSDC will also lay down the proportion of marks for Theory, Practical, Project, and Viva for each Element.
3. The assessment for the theory part will be based on the knowledge bank of questions created by the LSSSDC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. LSSSDC as assessment and awarding body will create unique evaluations for each assessment component i.e. theory, practical, project and viva for every student at each examination/training center based on this criterion.
6. Wherever any assessment component is not applicable/ feasible, the balance assessment components will be used to assess the candidate and accordingly the total marks will be calculated only for the applied



Qualification Pack

assessment component.

7. To pass the Qualification Pack, every trainee should score a minimum of 50%-70% of marks in each NOS (depending on NSQF Level) to successfully clear the assessment. In the case of a Govt funded program, the program guidelines will be overarching on the pass percentage rules.

8. In case of unsuccessful completion, the trainee may seek re-assessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0111.Ensure adherence to Environment, health and safety guidelines in production facility and GMP controlled areas by self and subordinates	30	55	-	15	100	15
LFS/N0129.Coordinate and communicate with supervisor/production chemist, research teams and auditors	32	33	17	18	100	15
LFS/N0572.Maintain experiment records ensuring data integrity and intellectual property rights	25	25	25	25	100	20
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	107	143	42	58	350	60

Elective: 1 API Synthesis & Medicinal Chemistry

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0516.Assist in development of new Active Pharmaceutical Ingredient (API) / Synthetic Product Development	40	20	20	20	100	20
LFS/N0573.Assist in Medicinal Chemistry research	40	40	10	10	100	20
Total	80	60	30	30	200	40

Elective: 2 Formulation Development

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0574.Assist in new Non Sterile formulation product development	40	40	10	10	100	20
LFS/N0575.Assist in aseptic product development	40	40	10	10	100	20
Total	80	80	20	20	200	40

Elective: 3 Biological Product Development

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0576.Assist in new biological product development	40	40	10	10	100	40
Total	40	40	10	10	100	40

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Elective: 4 Technology translation and Process Development

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0517.Assist in technology translation and process development	40	20	20	20	100	40
Total	40	20	20	20	100	40

Elective: 5 Technology transfer

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0577.Assist in technology transfer	40	20	20	20	100	40
Total	40	20	20	20	100	40

Elective: 6 Analytical - Mass Spectrometry

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0578.Conducts research and analysis using mass spectrometry techniques to solve complex problems, develop new products, or improve existing ones.	40	40	10	10	100	40
Total	40	40	10	10	100	40

Elective: 7 Analytical - NMR

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0579. Conducts research and analysis using Nuclear Magnetic Resonance techniques to solve complex problems, develop new products, or improve existing ones.	40	40	10	10	100	40
Total	40	40	10	10	100	40

Optional: 1 Innovation driven Entrepreneurship

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N0126. Establish innovation led enterprise and perform various entrepreneurial activities to run the regulated business operations	30	50	12	8	100	20
LFS/N0127. Maintain the Critical Business documents as Innovation led entrepreneur	23	48	17	12	100	20
LFS/N0128. Product commercialization and Intellectual Property Rights Management	40	40	10	10	100	20
Total	93	138	39	30	300	60

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
GMP	Good Manufacturing Practice
WHO	World Health Organization
ICH	International Council for Harmonisation
EHS	Environment Health and Safety
SOP	Standard Operating Procedure
SOP	Standard Operating Procedure
QA	Quality Assurance
EHS	Environment Health and Safety
POSH	Prevention of Sexual Harassment
PwD	People With Disability
ALCOA	Attributable, Legible, Contemporaneous, Original and Accurate
SOP	Standard Operating Procedure
LNB	Laboratory Notebook
eLNB	Electronic Laboratory Notebook
ALCOA	Attributable, Legible, Contemporaneous, Original, Accurate
NPD	New Product Development
API	Active Pharmaceutical Ingredient
PPE	Personal Protective Equipment
GRP	Good Research Practices
GLP	Good Laboratory Practices
NPD	New Product Development

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PPE	Personal Protective Equipment
GRP	Good Research Practices
GLP	Good Laboratory Practices
PPE	Personal Protective Equipment
GRP	Good Research Practices
GLP	Good Laboratory Practices
MSDS	Material Safety Data Sheet
NPD	New Product Development
MSDS	Material Safety Data Sheet
PPE	Personal Protective Equipment
GRP	Good Research Practices
GLP	Good Laboratory Practices
NPD	New product Development
PPE	Personal Protective Equipment
MSDS	Material Safety Data Sheet
GRP	Good Research Practices
GLP	Good Laboratory Practices
QC	Quality Control
CAQ	Critical Quality Attributes
CPP	Critical Process Parameters
DMF	Drug Master File
APIMF	API Master File
QC	Quality Control
CQA	Critical Quality Attributes
CPP	Critical Process Parameters
DMF	Drug Master File

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APIMF	API Master file
SOP	Standard Operating Procedures
NMR	Nuclear Magnetic Resonance
GDP	Good Documentation Practices
GDP	Good Documentation Practices
SOP	Standard Operating Procedure
SOP	Standard Operating Procedure
QA	Quality Assurance
EHS	Environment Health and Safety
POSH	Prevention of Sexual Harassment
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ALCOA	Attributable, Legible, Contemporaneous, Original and Accurate
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SOP	Standard Operating Procedure

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.